

Assignment 1: The Flatland Challenge

In the combined problems in this assignment (Part A and B), your job will be to schedule a set of trains through a railway network.

You will need to coordinate every train from its starting station to its destination as quickly as possible. As there can be many trains moving at the same time, you need to guarantee that each path is collision-free.

The assignment has three questions in increasing order of difficulty. The amount of points relative to each question is stated in the question heading. A total passing grade is 50%.

There are **two separate submission points** in this assignment.

Part A (question 1 only) will be due at an earlier date. Your job is to make a valid submission to the contest server by this due date (see the *FIT5222 Flatland Challenge Submission Instructions* for submission details). The marks you receive in this submission will be used for marking purposes for Q1. All points for Q1 can *only* be obtained by the Part A submission deadline.

The rest of the assignment, Part B, requires answering questions 2 and 3 as well as submitting a report detailing your approach and numerical results. Remember, each question in this assignment builds on top of the previous - it is important that you get started early and ask questions when you get stuck. Do not leave this assignment until the last minute!

Be sure to watch the introductory video on Moodle and to read the Introduction to Flatland documentation which we have prepared for you. Both are available from Moodle.

You must update your flatland code and installation before starting the assignment:

- **Under flatland folder**
- *git pull*
- *python setup.py install*
- **Now, run command *python -m pip list* , you should see flatland-rl version is updated to 2.2.4**
- **(Hint, use *python3* instead of *python* if *python3* points to the right one on your machine)**

Instructions to get code base (in **assignment1_2025 branch of piglet repo) for assignment is in the Introduction to Flatland documentation.**

Part A

Question 1 only, total 15 points, due **Sunday, 24 August 2025, 11:55 PM (end of Week 4)**.

Question 1: Warm up (15/150 points)

You are given start and target locations, one at a time. Your job is to route each train independently from all the rest. In this question collisions are not possible and there is no time dimension.

For this question, you need to implement a successor function for the Flatland domain. You also need to choose an algorithm to help you find paths. You are free to use any of the techniques we have discussed in the lectures, that you have read about in the literature or can write your own new approach.

Your solution will be evaluated on 40 evaluation instances (only staff have these instances) with a 2 hours time limit. Each instance has 1 agent. We will calculate a score, p_{score} , based on your sum of individual costs (SIC); for agents who did not arrive at their goal location, their cost will be $T_{max} = 8 * (width + height)$ of the map):

$$p_{score} = (completed_agents_SIC + T_{max} * failed_agents) / total_agents$$

We'll compare your solution to an optimal solution implemented by teaching team and calculate your score using following method:

$$f_{score} = p_{staff} / p_{score}$$

where p_{staff} is the p score of staff implementation. Generally, a good solution has small p score and large f score.

Your final score will be $f_{score}^i \times 15$ (since there are 15 points available for this question).

Marks for this question are available *only* by submission in Part A (i.e., no marks can be retroactively obtained when submitting Part B).

Submission

Part A submission to the contest server is due on ~~Sunday, 24 August 2025, 11:55 PM (end of Week 4)~~. **Extended to: Sunday 31st August 2025, 11:55 PM (end of Week 5).**

- Follow the ***FIT5222 Flatland Challenge Submission Instructions*** to submit your code to the contest server.

Part B

Questions 2, 3, and the report for a total of 135 points, due **Friday, 12 September 2025, 11:55 PM (end of Week 7)**.

Question 2: Basic scheduling (25/150 points)

You are given start and target locations, one at a time, as well as a set of existing paths for trains that are already moving. Your job is to route each train individually while avoiding collisions with all the rest. You are free to use any of the techniques we have discussed in the lectures, that you have read about in the literature or can write your own new approach.

For this question, you need to modify your successor function to account for time. In addition, there might be the situation that the search algorithm failed to find a feasible solution as dynamic obstacles block all possible paths. Just return an empty list in this case.

Furthermore, each action and each location for every computed plan need to be collision-free.

Your solution will be still evaluated on 56 instances with a 2 hours time limit and your score in this question will again be computed as the sum of individual path costs (SIC) and compared to the best solution from students (and staff)!

Your f_{score} will be computed in the same way as for Question 1. But there are some differences:

- Here we compute f_{score} for each instance.
- Each instance contains multiple agents.
- Your final points will be $\sum_i^n f_{score}^i \div 56 \times 25$, where i is instance id.

There are up to 25 points available for this question.

Question 3: Challenge (60/150 points)

You are given sets of start and target locations at the same time. Your job is to route all the trains simultaneously in a way that is collision-free. **But, each agent has an expected arrival time, late arrival will result in a penalty.**

You are free to use any of the techniques we have discussed in the lectures, that you have read about in the literature or can write your own new approach.

Now, as all agents are under your control, you need to make all agents reach their goal locations.

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In this question agents may run into malfunctions during execution. The evaluator will call the replan function when a new malfunction occurs. Implement the replan function to properly handle malfunction. Refer to the “Introduction to Flatland” document for details about malfunction and replan function.

Your solution will be still evaluated on 56 instances with different difficulty levels in 2 hours and your score in this question will again be computed as the sum of individual path costs (SIC) and compared to the best solution from students (and staff)!

Your f_{score} computation is similar to Question 1. But there are some differences:

- $p_{score} = (completed_agents_SIC + penalty + T_{max} * failed_agents) / total_agents$
- The *penalty* is $2 * total_delayed_timesteps$.
- Each instance contains multiple agents.
- Each instance's f_{score} will refer to a baseline implementation p_{base} and an advanced implementation p_{adv} (or best student solution which ever is better):

$$f_{score} = \min(0.5 * \frac{p_{base}}{p_{score}}, 0.5) + \max(0.5 * \frac{p_{base} - p_{score}}{p_{base} - p_{adv}}, 0)$$

- $\sum_i^n f_{score}^i \div 56 \times 60$, where i is instance id.

There are up to 60 points available for this question.

Report (50/150 points)

You are required to prepare a comprehensive report that explains your approach to each of the questions in detail. The report contributes **50 marks** to your final grade.

Your submission must include:

- A written explanation of the methods you used,
- Justification for choosing those methods, and
- A critical discussion supported by relevant evidence (e.g., pseudocode, images, graphs, tables, etc.).

Reports should be limited at 3000 words (*not* including supporting evidence such as pseudocode, images, graphs, and tables).

A *suggestion* for the each question division is:

- **Q1:** 800 words (exceeding this limit will result in penalties).
- **Q2:** 1000 words (exceeding this limit will result in penalties).
- **Q3:** 1200 words (exceeding this limit will result in penalties).

Bonus Points

We will issue bonus points for students implementing multiple approaches for the same problem, and/or for implementations of algorithms from the scientific literature that are not discussed during tutorials/lectures (e.g., from one of the recommended papers or elsewhere). The size of the bonus depends on how ambitious the implementation is, its effectiveness and the quality of the writeup.

BONUS POINTS CAN MAKE THE DIFFERENCE BETWEEN 'D' and 'HD'

Grades

The total points for the assignment is 150. Part A is worth 15 marks, and Part B is worth 135 marks. A passing grade is 50% of the total points, which is 75 points.

Submission

The assignment is due on **Friday, 12 September 2025, 11:55 PM (end of Week 7)**. You must submit your work in the following two places.

1. Submit your code to the Contest Server:

- Follow the ***FIT5222 Flatland Challenge Submission Instructions*** to submit your code to the contest server.
- Your code will be auto marked on the contest server. You **cannot** receive marks for your code submission if you do not submit to the contest server.
- You must submit your code using your Monash credentials. You **cannot** receive marks for code submitted via a personal account.

2. Submit your code and report to Moodle:

1. Your implementation source code, in a single directory called "src" (you can copy everything in the piglet folder to "src"). Zip the codes directory with file name last_name_student_id_flatland.zip. (For example, Chen_123456_flatland.zip)
2. A scientific report describing your approaches as a single pdf file. Name the pdf as last_name_student_id_report.pdf. (For example, Chen_123456_report.pdf)
3. Submit the two files separately to the Moodle assignment submission page. (**Do not zip the pdf report, and make sure your PDF has selectable text.** Moodle needs a searchable PDF to generate TurnItIn report). Failure to submit a correctly formatted report may incur a penalty.

Generative AI

There are no restrictions on use of generative AI for this assessment task, but students should accomplish their assessments individually. Students **MUST** acknowledge any use of generative AI with an AI statement in their report.

See the Report Rubric Below!

Report Marking Rubric

Criteria	N 0%-49%	P 50%-59%	C 60%-69%	D 70%-79%	HD 80%-100%
Approach Description (15 points)	Incomplete or insufficient description of the approach. Does not demonstrate sufficient understanding of the implemented approach and related algorithms.	There is a description of the approach, but the approach is generally unclear. Little or no evidence to support the discussion. The discussion demonstrates a limited understanding of the implemented approach and related algorithms.	There is a description of the approach that can be mostly understood, except in respect of small details. There is limited evidence and a limited argument justifying the choice of approach. The discussion demonstrates a baseline understanding of the implemented approach and related algorithms.	There is a description of the approach that can be fully understood. There is a clear but somewhat flawed argument justifying the methodology. The approach is supported by evidence that is somewhat relevant. The discussion demonstrates good understanding of the implemented approach and related algorithms, but with minor issues.	There is a description of the approach that can be fully understood. The quality of the justification of the methodology is supported by strong evidence. The discussion demonstrates excellent understanding of the implemented approach and related algorithms.
Theoretical Analysis (5 points)	Discussion is minimal, superficial, or absent. No evidence of algorithmic analysis. Demonstrates little to no understanding of theoretical concepts.	Some attempt at algorithmic analysis is present, but is incomplete or inaccurate. Provides a basic description of theoretical concepts, but demonstrates narrow understanding.	Discussion covers key aspects of algorithmic analysis but lacks depth or precision. Demonstrates a sound understanding of relevant theoretical concepts, but with some key mistakes.	Provides an accurate overview algorithmic analysis of time & space complexity, completeness, and/or optimality. Demonstrates good understanding of theoretical concepts with few mistakes.	Provides a complete and rigorous algorithmic analysis, (time, space, completeness, optimality..). Demonstrates a comprehensive and insightful understanding of theoretical concepts.
Experiment Discussion (10 points)	Provides little or no discussion of experimental work.	Provides a basic discussion of experimental results	Presents a sound discussion of experiments with supporting numerical	Provides a strong and well-supported discussion of experiments. Numerical	Demonstrates a comprehensive and insightful discussion of

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	Results are missing, incomplete, or unsupported by evidence. No numerical details are presented, and efficiency or implementation performance is not addressed. Work shows major flaws or misunderstanding of the experimental process.	with limited evidence. Some numerical details may be included but are incomplete or lack context. Mentions efficiency of implementation in vague terms, without meaningful analysis or comparison. Discussion shows only minimal engagement with experiments performed.	details. Results are generally clear and partially analysed for efficiency. Some attempt to evaluate implementations against benchmarks or references is made, but analysis may be superficial or inconsistent. Shows adequate understanding of the experimental process.	evidence is clearly presented and used to evaluate efficiency of implementations. Comparative analysis against standard benchmarks and/or appropriate reference algorithms is evident and accurate. Discussion demonstrates critical engagement, though some aspects may lack full depth.	experiments. Numerical results are detailed, accurate, and fully integrated into the analysis. Efficiency of implementations is rigorously assessed on standard benchmarks and compared effectively to appropriate reference algorithms. Evidence is consistently used to support claims, and discussion reflects exceptional depth, precision, and critical evaluation.
Reflections (10 points)	Lacks critical thinking and original insights. Discussion of the strengths and weaknesses of the implemented approach are absent or irrelevant.	Limited critical thinking and few original insights. Conclusions are vague or poorly articulated, but touch upon some strengths and weaknesses of the implemented approach.	Demonstrates basic critical thinking with limited insights. Provides some conclusions, but misses key implications relevant to the implemented approach.	Shows good critical thinking with some reflections on the strengths and weaknesses of implemented approaches. Provides clear discussion of implications, but misses some key points.	Demonstrates strong critical thinking and provides insightful reflections on the strengths and weaknesses of implemented approaches. Clearly articulates the implications of the findings.
Communication skills (10 points)	Hard to follow with no clear narrative. Inadequate or no separation of discussion text into	The writing has a tenuously logical narrative. Some attempt at the expected structural elements (e.g. Intro, conclusion).	The text has a clear logical narrative and expected structural elements (e.g. intro, conclusion).	The writing is well composed and has a clear and logical narrative and is well structured. Writing is generally	The writing is very well composed and has a very clear and logically-formed narrative as a whole. Writing is accurate and

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	coherent sections. Writing is not accurate or articulate. Inadequate supporting materials. Inadequate or missing referencing.	Writing is not accurate or articulate most of the time. The document has few supporting materials (tables, images, pseudo-code) . The student has attempted to undertake citing and referencing with frequent errors.	Writing is not accurate or articulate most of the time. There are some supporting materials (tables, images, pseudo-code) but not well integrated with the rest of the text. The student follows the requirements for citing and referencing, with some notable errors.	accurate and articulate. The document has appropriate supporting materials that are well integrated with the rest of the text. The student follows the requirements for citing and referencing, with some minor errors.	articulate. The document is expertly structured in the style of a scientific report, including appropriate supporting materials that clearly improve the quality of associated discussion. The student follows the requirements for citing and referencing.
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